

Online Supplement to “Tilted Cell Moments: A One-Pass Exact Newton Method for Generalized Linear Models at Scale”

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This supplement develops the companion theory and additional methodology for the framework of the main paper: global convergence and local rates for the nonlinear alternating-projections absorption of high-dimensional fixed effects (Sections S1, with the $G \geq 3$ rate theory in Section S1.4); asymptotic theory in fixed, diverging, and proportional-dimension regimes (Section S2); further extensions—families beyond Poisson at high dimension, penalized estimation, and Firth’s bias-reducing adjustment (Section S3); simulation studies validating each result (Section S4); and all proofs. Notation follows the main paper; numbered objects here carry the prefix S.

Notation recalled

Observations $i = 1, \dots, N$ carry responses y_i from an exponential dispersion family with cumulant b , individual-level covariates $\mathbf{x}_i \in \mathbb{R}^p$, cell-level covariates $\mathbf{w}_{j(i)} \in \mathbb{R}^k$ constant within the cells $j = 1, \dots, J$ of the categorical cross-classification, and offsets o_i ; $\boldsymbol{\beta} = (\boldsymbol{\gamma}, \boldsymbol{\delta})$ collects the individual- and cell-level coefficients, and for canonical links $\omega_i = b''(\eta_i)$. The tilted cell moments of main-paper Theorem 1 are

$$M_j^{(0)} = \sum_{i \in j} \omega_i, \quad \mathbf{m}_j = \sum_{i \in j} \omega_i \mathbf{x}_i, \quad \mathbf{Q} = \sum_{i=1}^N \omega_i \mathbf{x}_i \mathbf{x}_i', \quad (\text{S1})$$

from which the exact score and information are assembled at $O(N(p^2 + p)) + O(J(k^2 + kp))$ per Newton or Fisher-scoring iteration. Algorithm S1 below denotes the absorbed-factor scheme analyzed here: cyclic closed-form profile updates of the absorbed effects alternated with damped tilted-moment Newton steps for $\boldsymbol{\beta}$.

Algorithm S1 Tilted-moment Newton with absorbed factors (nonlinear block Gauss–Seidel)

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1: Initialize  $\beta_0, \alpha_0 = \mathbf{0}$ ; drop observations in absorbed groups with no events (Section S1.5).
2: repeat
3:   repeat ▷ inner loop: exact block maximization over  $\alpha$ 
4:     for  $g = 1, \dots, G$  do
5:       update  $\alpha^{(g)}$  by the closed-form profile rule (S3)
6:     end for
7:   until  $\max_g \|\Delta \alpha^{(g)}\|_\infty < \tau_{\text{in}}$ 
8:   accumulate tilted moments (S1) at  $(\beta_t, \alpha_t)$  in one microdata pass
9:    $\beta_{t+1} \leftarrow \beta_t + \lambda_t \mathbf{F}^{-1} \mathbf{U}$  with  $\lambda_t \in \{1, \frac{1}{2}, \frac{1}{4}, \dots\}$  chosen by backtracking (Armijo) on
      $\ell(\cdot, \hat{\alpha}(\cdot))$ 
10: until  $\max_\ell |\Delta \beta_\ell| / (1 + |\beta_\ell|) < \tau$ 
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S1 High-dimensional fixed effects and nonlinear alternating projections

S1.1 Profile formulation

Beyond the cell design, applications routinely require one or more fixed-effect factors whose level counts preclude even cell-level treatment: firms, hospitals, zip-code–year interactions. Augment the linear predictor with G absorbed factors,

$$\eta_i = \gamma' \mathbf{x}_i + \delta' \mathbf{w}_{j(i)} + \sum_{g=1}^G \alpha_{g(i)}^{(g)} + o_i, \quad (\text{S2})$$

where factor g has J_g levels with level map $g(i)$ and effect vector $\alpha^{(g)} \in \mathbb{R}^{J_g}$, and $J_1 + \dots + J_G$ may be of order N . The full parameter is $\theta = (\beta, \alpha)$ with $\beta = (\gamma, \delta)$. The design has a known null space $\mathcal{K} \subset \mathbb{R}^{q+\sum_g J_g}$ (at least the G dimensions of shifts between the factors' levels and the intercept, more if the factor graph is disconnected); all statements below are on the quotient $\mathbb{R}^{q+\sum J_g} / \mathcal{K}$, realized in practice by a normalization. The estimator of interest is the MLE of β with α concentrated out: the profile likelihood $\ell_p(\beta) = \max_\alpha \ell(\beta, \alpha)$.

For the Poisson family the inner maximization has a closed-form coordinate solution: given all other parameters, the optimal effect of level a of factor g is

$$\alpha_a^{(g)} = \log \frac{\sum_{i:g(i)=a} y_i}{\sum_{i:g(i)=a} \exp(\eta_i - \alpha_a^{(g)})}, \quad (\text{S3})$$

the log of the ratio of observed to expected-at-zero-effect events, which is the multiplicative-update rule used by FENmlm/fixest (Bergé, 2018) and, in IRLS-linearized form, by ppmlhdfe (Correia et al., 2020). Cycling (S3) over factors is a nonlinear block Gauss–Seidel scheme; combined with tilted-moment Newton steps for β it yields Algorithm S1.

Algorithm S1 is, up to the tilted-moment acceleration of the β -step, the algorithm used across the applied HDFE–GLM software ecosystem. Its convergence status splits cleanly in two. The *inner* loop is the method of alternating projections, whose convergence is classical—von Neumann for two subspaces, generalized by Halperin (1962), with the Kaczmarz sweep (Kaczmarz, 1937) as the coordinate case—and which Gaure (2013) introduced into fixed-effects estimation on exactly this foundation; Correia et al. (2020) prove convergence of the linear inner solves they employ, and Deutsch and Hundal (1997) supply rates. For the *combined* nonlinear cycle—profile updates of the absorbed effects alternated with Newton/scoring steps on the GLM likelihood—the literature offers simulation evidence (Guimarães and Portugal, 2010; Bergé, 2018; Correia et al., 2020) but no unified formal theorem. The next two subsections supply that missing statement: Theorem S1 (main-paper Theorem 2) proves global convergence of the combined inner–outer iteration, and Theorems S2–S3 characterize its rate.

S1.2 Global convergence

I impose:

Assumption S1. *The link is canonical and b is strictly convex and twice continuously differentiable on $\Theta = \mathbb{R}$.*

Assumption S2. *The MLE exists: $\sup_{\theta} \ell$ is attained at some θ^* (unique modulo $\mathcal{K}$).*

Assumption S3. *(i) Each absorbed group contains at least one observation with y_i in the interior of the convex hull of the support of the family (for Poisson and binomial: at least one $y_i > 0$, and for binomial also one $y_i < n_i$), so that each one-dimensional block likelihood in (S3) attains a finite maximum; (ii) the β -block information $\mathbf{F}_{\beta\beta}(\theta)$ is nonsingular on the initial superlevel set $\{\ell \geq \ell(\theta_0)\}$.*

Assumption S2 is a data condition—absence of separation—with a long literature of verifiable characterizations (Haberman, 1974; Wedderburn, 1976; Albert and Anderson, 1984; Santos Silva and Tenreiro, 2010; Correia et al., 2019); Section S1.5 discusses enforcement and the Firth alternative when it fails. Assumption S3(i) is enforced constructively by the drop rule of Algorithm S1, mirroring standard practice (Correia et al., 2020).

Theorem S1 (Global convergence of nonlinear alternating projections; main-paper Theorem 2). *Under Assumptions S1–S3, the iterates $\theta_t = (\beta_t, \alpha_t)$ of Algorithm S1 satisfy: (i) $\ell(\theta_t)$ is nondecreasing and converges to $\sup \ell = \ell(\theta^*)$; (ii) $[\theta_t] \rightarrow [\theta^*]$ in the quotient $\mathbb{R}^{q+\sum J_g}/\mathcal{K}$; in particular $\beta_t \rightarrow \beta^*$ whenever the normalization places the flat directions in the absorbed blocks; and (iii) the same conclusions hold if the β -update is replaced by exact block maximization, and for any cyclic or essentially-cyclic ordering of the blocks; and (iv) Assumption S1 may be weakened to the requirement that the link be twice continuously differentiable with each*

per-observation log-likelihood contribution strictly concave in its linear predictor (a log-concave link; Pratt, 1981), provided the β -step is an Armijo-damped ascent step with a symmetric preconditioner whose eigenvalues are bounded within $[\lambda, \bar{\lambda}] \subset (0, \infty)$ on the initial superlevel set—Fisher scoring in particular.

Part (iv) settles the scope of the guarantee beyond canonical links. Log-concavity of the per-observation contribution in η_i is exactly the condition under which the proof’s three ingredients survive—concavity of ℓ in θ , unique finite block maximizers, and sufficient ascent of a preconditioned gradient step—and it is satisfied well beyond the canonical family: Pratt (1981) characterizes the log-concave links, which include the gamma family with log link (each contribution $-\nu\{y_i e^{-\eta_i} + \eta_i\}$ has second derivative $-\nu y_i e^{-\eta_i} < 0$), the binomial probit, and the binomial complementary log-log. For the gamma-log pair the absorbed blocks even retain a closed-form update, $\alpha_a^{(g)} = \log\{m_a^{-1} \sum_{i:g(i)=a} y_i \exp(-(\eta_i - \alpha_a^{(g)}))\}$ with m_a the group size, so Algorithm S1 applies unchanged. For links outside this class the log likelihood need not be concave in θ and the guarantee weakens to what smooth ascent methods generically provide: monotone likelihood ascent with stationary limit points, without a global claim; I state this boundary explicitly since it is where the folk confidence in the algorithm stops being a theorem.

The proof (Appendix S-A) proceeds by establishing that strict concavity on the quotient plus attainment forces compact superlevel sets (Lemma S2); monotonicity then confines the iterates to a compact set, every limit point is shown to be a blockwise maximizer using the uniqueness of the block updates and the sufficient-ascent property of the damped Newton step, and for smooth concave functions blockwise maximizers are global maximizers. The argument is in the tradition of block-coordinate methods (Bertsekas, 1999; Tseng, 2001); what requires care is (a) the quotient structure (the likelihood is not coercive in the natural parametrization, only modulo $\mathcal{K}$), (b) the inexact β -block (a damped Newton step, not an exact maximizer), and (c) dispensing with any compactness or boundedness assumption on the parameter space, which would be false here and is where informal appeals to standard results break down.

Theorem S1 has a practical corollary worth stating: since every accumulation point of the algorithm is *the* MLE, the familiar diagnostics of the linear-projection literature (sensitivity of estimates to tightened tolerances, discrepancies across implementations) reflect finite-iteration error only, not multiplicity; and step-halving on the profile objective—re-solving the inner loop at every trial point, as Algorithm S1 specifies—is guaranteed to terminate at each outer iteration.

S1.3 Local rate: Friedrichs angles and a bipartite spectral gap

Convergence speed is governed by the geometry of the absorbed subspaces. Let $\Omega^* = \text{diag}(\omega_i^*)$ collect the information weights at the optimum, let $V_g = \text{span}\{\text{indicator columns of factor } g\} \subset \mathbb{R}^N$, and equip $\mathbb{R}^N$ with the inner product $\langle u, v \rangle_{\Omega^*} = u' \Omega^* v$. Recall that the *Friedrichs angle*

$\theta_F \in (0, \pi/2]$ between closed subspaces V_1, V_2 with intersection V_0 has cosine

$$c_F = \sup \left\{ \frac{\langle u, v \rangle_{\Omega^*}}{\|u\|_{\Omega^*} \|v\|_{\Omega^*}} : u \in V_1 \cap V_0^\perp, v \in V_2 \cap V_0^\perp, u, v \neq \mathbf{0} \right\} < 1.$$

Theorem S2 (Local linear convergence). *Let Assumptions S1–S3 hold and let b be three times continuously differentiable. (i) The inner iteration of Algorithm S1 at fixed β (the map $T : \alpha \mapsto \alpha^+$ of one Gauss–Seidel sweep of exact block maximizations) is continuously differentiable in a neighborhood of its fixed point $\hat{\alpha}(\beta)$, with Jacobian equal to the block Gauss–Seidel iteration matrix of the α -block information $\mathbf{F}_{\alpha\alpha}$ at the fixed point; consequently the inner iteration converges locally R -linearly with rate $\rho_{\text{GS}} = \rho((\mathbf{D} + \mathbf{L})^{-1}\mathbf{L}') < 1$ on the quotient, where $\mathbf{F}_{\alpha\alpha} = \mathbf{D} + \mathbf{L} + \mathbf{L}'$ is the block decomposition into diagonal and strictly lower/upper block-triangular parts. (ii) For $G = 2$ absorbed factors,*

$$\rho_{\text{GS}} = c_F^2,$$

the squared Friedrichs cosine between V_1 and V_2 in the Ω^ inner product; the same rate governs the weighted alternating-projections demeaning of any working vector, by the theorem of Aronszajn (1950) and Kayalar and Weinert (1988). (iii) The outer iteration (full cycles of Algorithm S1 with exact β -maximization) converges locally R -linearly with rate at most $\rho(\text{block GS operator of } \mathbf{F}(\theta^*) \text{ in the ordering } (\alpha^{(1)}, \dots, \alpha^{(G)}, \beta)) < 1$.*

Corollary S1 (Bipartite spectral gap). *For $G = 2$, define the weighted biadjacency matrix $\mathbf{B} \in \mathbb{R}^{J_1 \times J_2}$ by $B_{ab} = \sum_{i: g_1(i)=a, g_2(i)=b} \omega_i^*$, and let $\mathbf{D}_1 = \text{diag}(\mathbf{B}\mathbf{1})$, $\mathbf{D}_2 = \text{diag}(\mathbf{B}'\mathbf{1})$. Then*

$$c_F = \sigma_2(\mathbf{D}_1^{-1/2} \mathbf{B} \mathbf{D}_2^{-1/2}),$$

the second-largest singular value of the normalized biadjacency (transition) kernel. Equivalently, $1 - c_F^2$ is the spectral gap of the random walk on the weighted bipartite graph of the two factors, and $c_F < 1$ if and only if that graph is connected—the classical identification condition for two-way fixed effects (Abowd et al., 1999; Gaure, 2013).

Corollary S1 turns the folklore statement “correlated fixed effects make demeaning slow” into a computable quantity: before iterating at all, σ_2 of a $J_1 \times J_2$ sparse matrix (a few Lanczos iterations) predicts the per-sweep contraction factor exactly. It also gives sharp intuition for the pathologies: nearly disconnected mobility graphs—workers who rarely switch firms, patients who rarely switch hospitals—have $\sigma_2 \uparrow 1$ and predictably slow demeaning, while balanced crossed designs have small σ_2 and converge in a handful of sweeps. Section S4.1 verifies the prediction to three decimals across two orders of magnitude of $1 - c_F^2$.

S1.4 Three or more absorbed factors

Administrative designs routinely absorb three or more factors—worker, firm, and region—year, say—and for $G \geq 3$ the linear alternating-projections literature offers norm bounds but no clean angle characterization of the asymptotic rate (Deutsch, 2001; Badea et al., 2012). The following theorem resolves the practical question for the present setting: the exact local rate for any G is a computable spectral quantity determined by the *pairwise* weighted contingency structure of the factors alone, so the bipartite intuition of Corollary S1 does not break down at $G \geq 3$ —it generalizes, with the pair graph replaced by a G -partite sweep operator. For each pair $g \neq h$ define the weighted cross-tabulation $\mathbf{B}_{gh} \in \mathbb{R}^{J_g \times J_h}$, $(\mathbf{B}_{gh})_{ab} = \sum_{i:g(i)=a, h(i)=b} \omega_i^*$, and the level totals $\mathbf{D}_g = \text{diag}(\mathbf{B}_{gh} \mathbf{1})$ (independent of h).

Theorem S3 (Exact rate and certificates for $G \geq 2$ factors). *Let the assumptions of Theorem S2 hold and let $\mathcal{F} = \mathbf{F}_{\alpha\alpha}(\boldsymbol{\theta}^*)$ denote the absorbed-block information on the level space $\mathbb{R}^{J_1 + \dots + J_G}$.*

- (i) (Pairwise sufficiency.) *$\mathcal{F}$ has diagonal blocks $\mathbf{D}_g$ and off-diagonal blocks $\mathbf{B}_{gh}$; hence the local rate ρ_{GS} of Theorem S2(i) is a function of the pairwise weighted cross-tabulations only, and can be evaluated without access to the microdata once the $G(G-1)/2$ tables are accumulated (one $O(N)$ pass).*
- (ii) (Exact computable rate.) *Write $\mathcal{F} = \mathcal{D} + \mathcal{L} + \mathcal{L}'$ with $\mathcal{D} = \text{diag}(\mathbf{D}_1, \dots, \mathbf{D}_G)$ and $\mathcal{L}$ strictly block lower triangular, and let $\mathbf{M} = -(\mathcal{D} + \mathcal{L})^{-1} \mathcal{L}'$ be the sweep operator. Then the eigenspace of $\mathbf{M}$ for the eigenvalue 1 is exactly $\text{null}(\mathcal{F})$ (the flat shift directions), this eigenvalue is semisimple, and*

$$\rho_{\text{GS}} = \max \{ |\lambda| : \lambda \in \text{spec}(\mathbf{M}), \lambda \neq 1 \} < 1 \iff \text{null}(\mathcal{F}) = \text{flats} \iff \text{the multi-way design is identified.}$$

Because the eigenvalue 1 is semisimple, successive differences of the sweep iteration decay at exactly ρ_{GS} , so the rate is computable by power iteration on the level space at $O(\sum_{g < h} \text{nnz}(\mathbf{B}_{gh}))$ per step.

- (iii) (A priori certificate.) *With c_g the Friedrichs cosine, in the Ω^* inner product, between V_g and $V_{g+1} + \dots + V_G$ for $g = 1, \dots, G-1$,*

$$\rho_{\text{GS}} \leq \|\text{one demeaning sweep}\|_{\Omega^*} \leq \left(1 - \prod_{g=1}^{G-1} (1 - c_g^2) \right)^{1/2},$$

each c_g again being a level-space quantity (the largest non-unit generalized singular value between factor g and the pooled design of the later factors); $c_g < 1$ for all g if and only if the design is identified.

- (iv) *For $G = 2$, (ii) reduces to $\rho_{\text{GS}} = c_F^2 = \sigma_2^2$ of Corollary S1, and the bound in (iii) to c_F .*

Three comments. First, part (ii) is the operative solution for practice: the exact rate is available *before iterating on the microdata*, from cross-tabulations that cost one linear pass, by power iteration on vectors of length $\sum_g J_g$; the reference implementation exposes this as `gs_spectral_rate`, and Section S4.1 shows it matching observed contraction factors to three decimals for $G = 3$ exactly as for $G = 2$. Second, part (iii) gives the interpretable geometry: each successive factor slows the sweep according to its angle to the span of the factors after it, the product-of-sines form being the multi-subspace bound of Smith et al. (1977) and Deutsch and Hundal (1997) transported through the complement duality; the bound is attained-in-order for $G = 2$ but is conservative for $G \geq 3$ (Table S2), which is precisely why (ii), not an angle formula, is the right object to compute. The sharp one-sweep *energy-norm* contraction for general G is characterized abstractly by the identity of Xu and Zikatanov (2002) for multiplicative subspace-correction methods—the demeaning sweep is exactly such a method—but that identity’s constant is itself a variational quantity no easier than (ii), reinforcing the same conclusion. Third, the identification equivalence in (ii) generalizes the connectedness criterion of Abowd et al. (1999) and Gaure (2013): for $G \geq 3$, pairwise connectedness of all factor pairs is necessary but not sufficient for $\rho_{\text{GS}} < 1$ on the quotient; the spectral condition is exactly right, and the power iteration returns $\rho_{\text{GS}} = 1$ (to tolerance) when identification fails, making it a cheap diagnostic as well as a rate predictor.

S1.5 Separation

Assumption S2 fails in the maximum-likelihood sense when a hyperplane in covariate space separates events from non-events: components of $\hat{\theta}$ diverge and the supremum is not attained (Albert and Anderson, 1984; Santos Silva and Tenreiro, 2010). In the absorbed blocks the failure mode is transparent—a group with no events has $\hat{\alpha}_a = -\infty$ by (S3)—and Algorithm S1 drops such groups a priori, after which Assumption S3(i) holds by construction; this mirrors Correia et al. (2019), who treat detection in full generality. Separation induced by the β -block cannot be dropped away and surfaces as non-convergence. Section S3.3 develops the principled remedy—Firth’s adjusted score inside the tilted accumulation—and Section S4.3 quantifies its behavior.

S2 Asymptotic theory

Because Theorem 1 of the main paper makes the tilted-moment estimator *equal* to the MLE, its asymptotic analysis is the analysis of the MLE in the relevant regime. I give results in three regimes, in increasing order of dimensionality. Throughout, θ_0 denotes the truth, expectations are under the model, and the triangular-array framework lets all design quantities depend on N .

S2.1 Fixed dimension

Assumption S4. (i) $q = p + k$ is fixed; covariates and offsets are uniformly bounded; (ii) $\lambda_{\min}(\mathbf{F}_N(\boldsymbol{\theta}_0)) \rightarrow \infty$ and $N^{-1}\mathbf{F}_N(\boldsymbol{\theta}_0) \rightarrow \mathbf{F}_\infty \succ \mathbf{0}$; (iii) $\boldsymbol{\theta}_0$ is interior to the parameter space and b is three times continuously differentiable.

Theorem S4 (Classical asymptotics). *Under Assumptions S1 and S4, with probability tending to one the MLE $\hat{\boldsymbol{\theta}}_N$ exists, is unique, and satisfies $\sqrt{N}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \phi \mathbf{F}_\infty^{-1})$; the estimator is asymptotically efficient, attaining the Cramér–Rao bound. If the conditional mean is correctly specified but the distribution is not, the same limit holds with sandwich variance $\mathbf{F}_\infty^{-1} \boldsymbol{\Sigma} \mathbf{F}_\infty^{-1}$, consistently estimated by the cell-accumulated sandwich of Section 4.4 of the main paper.*

The proof (Appendix S-B) is a specialization of Fahrmeir and Kaufmann (1985) plus the pseudo-ML results of Gouriéroux et al. (1984); I include it because the triangular-array statement with offsets and the exact form of the cell-accumulated sandwich are used repeatedly later.

S2.2 Diverging dimension: $q_N^3/N \rightarrow 0$

Register applications place $k_N = \dim(\boldsymbol{\delta})$ in the hundreds or thousands—period $\times$ region $\times$ education cells—so the fixed- q regime is a poor description. Let $q_N = p + k_N \rightarrow \infty$, write $\mathbf{F}_N = \mathbf{F}_N(\boldsymbol{\theta}_0)$, and define the leverage of observation i as $h_i = \omega_i \mathbf{a}_i' \mathbf{F}_N^{-1} \mathbf{a}_i$ (so $\sum_i h_i = q_N$).

Assumption S5. (i) Covariates and offsets are uniformly bounded, and the true linear predictors $\{\eta_{0i}\}$ lie in a fixed compact set; (ii) balanced leverage: $\max_i h_i \leq C q_N/N$ for a fixed constant C ; (iii) $q_N^3/N \rightarrow 0$; (iv) b is three times continuously differentiable with b'' bounded away from 0 and ∞ , and b''' bounded, on compact sets.

Condition (ii) is the natural design condition: for a pure cell design with balanced cells it holds with C close to 1, and it fails exactly when some cell is drastically undersized relative to the average—the same configurations for which practitioners distrust the asymptotics.

Theorem S5 (Diverging dimension). *Under Assumptions S1 and S5, with probability tending to one the MLE exists and satisfies*

$$\|\mathbf{F}_N^{1/2}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0)\| = O_p(\sqrt{q_N}),$$

and for any fixed-dimensional matrix of contrasts $\mathbf{C}_N' \in \mathbb{R}^{d \times q_N}$ with $\mathbf{C}_N' \mathbf{F}_N^{-1} \mathbf{C}_N \rightarrow \boldsymbol{\Sigma}_C \succ \mathbf{0}$,

$$(\mathbf{C}_N' \mathbf{F}_N^{-1} \mathbf{C}_N)^{-1/2} \mathbf{C}_N' (\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \phi \mathbf{I}_d).$$

The plug-in variance estimator $\mathbf{C}_N' \mathbf{F}_N(\hat{\boldsymbol{\theta}}_N)^{-1} \mathbf{C}_N$ is consistent in relative terms.

The proof (Appendix S-B.2) is self-contained and short by the standards of this literature, because concavity in the canonical parametrization substitutes for the empirical-process machinery needed in general M -estimation (Portnoy, 1988; He and Shao, 2000; Wang, 2011): a quadratic-minorization argument in the intrinsic norm delivers the rate, and a third-order Taylor control with the balanced-leverage condition delivers asymptotic linearity with remainder $o_p(1)$ exactly when $q_N^3/N \rightarrow 0$.

S2.3 The proportional regime $J/N \rightarrow c$: Poisson profile likelihood as conditional likelihood

The regime that motivates the proportional-dimension question is more extreme: one absorbed effect per cell with cells of *bounded* size, so the parameter count is a positive fraction of the sample size and Theorem S5 cannot apply ($q_N \asymp N$). This is the incidental-parameters regime of Neyman and Scott (1948); for smooth nonlinear panel models the profile MLE of the common parameter is typically inconsistent or biased at order $1/m$ for cell size m (Lancaster, 2000; Fernández-Val and Weidner, 2016). The Poisson family with log link is a distinguished exception, for a structural reason worth displaying: profiling and conditioning coincide.

Consider cells $j = 1, \dots, J$ of sizes $m_j \leq \bar{m} < \infty$, one free effect α_j per cell, individual-level covariates $\mathbf{x}_{ij}$, offsets o_{ij} , and $y_{ij} \sim \text{Poisson}(\exp(\alpha_j + \gamma' \mathbf{x}_{ij} + o_{ij}))$ independently. Let $D_j = \sum_i y_{ij}$.

Lemma S1 (Profile equals conditional). *For every γ with all D_j -positive cells,*

$$\ell_p(\gamma) = \max_{\alpha} \ell(\gamma, \alpha) = \sum_{j: D_j > 0} \log \mathbb{P}(y_{j1}, \dots, y_{jm_j} \mid D_j; \gamma) + \text{terms not involving } \gamma,$$

where the conditional law of $(y_{j1}, \dots, y_{jm_j})$ given D_j is multinomial with D_j trials and probabilities $\pi_{ij}(\gamma) = e^{\gamma' \mathbf{x}_{ij} + o_{ij}} / \sum_{i'} e^{\gamma' \mathbf{x}_{i'j} + o_{i'j}}$. Moreover ℓ_p is concave in γ , and its score and negative Hessian are the Schur complements of the tilted moments:

$$\mathbf{U}_p(\gamma) = \sum_j \sum_i (y_{ij} - D_j \pi_{ij}) \mathbf{x}_{ij}, \quad \mathbf{H}_p(\gamma) = \sum_j D_j \text{Var}_{\pi_j}(\mathbf{x}_j) = \mathbf{Q} - \sum_j \frac{\mathbf{m}_j \mathbf{m}_j'}{M_j^{(0)}}.$$

Lemma S1 is a fixed-effects analogue of the classical Poisson–multinomial connection; versions of the first display appear in the econometric panel literature (Andersen, 1970; Hausman et al., 1984; Blundell et al., 2002; Wooldridge, 1999), and the Schur-complement identity is what the tilted-moment machinery computes at $O(Np^2)$ per iteration (function `profile_poisson` in the accompanying code). Its consequence, however, seems not to have been developed as an explicit high-dimensional limit theorem with variance theory, which I now state. Conditional on covariates and cell effects, cells are independent; define the average conditional information $\bar{\mathbf{F}}_J(\gamma_0) = J^{-1} \sum_j \mathbb{E}[D_j \text{Var}_{\pi_j}(\mathbf{x}_j)]$.

Assumption S6. (i) Cell sizes satisfy $2 \leq m_j \leq \bar{m}$; $J/N \rightarrow c \in (0, 1)$; (ii) covariates and offsets are uniformly bounded and $\sup_j \alpha_j \leq C$, so cell means are bounded above (they may be arbitrarily small); (iii) $\bar{\mathbf{F}}_J(\gamma_0) \rightarrow \bar{\mathbf{F}} \succ \mathbf{0}$ as $J \rightarrow \infty$; (iv) γ_0 is interior to a fixed compact set Γ .

Theorem S6 (Proportional regime). Under Assumption S6: (i) the profile MLE $\hat{\gamma}_J$ is consistent; (ii)

$$\sqrt{J}(\hat{\gamma}_J - \gamma_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \bar{\mathbf{F}}^{-1}),$$

with no asymptotic bias, despite the number of nuisance parameters being a positive fraction of N ; (iii) both the profile information estimator $J \mathbf{H}_p(\hat{\gamma}_J)^{-1}$ and the cell-clustered sandwich estimator $J \mathbf{H}_p^{-1}(\frac{G}{G-1} \sum_j \mathbf{s}_j \mathbf{s}_j') \mathbf{H}_p^{-1}$, with $\mathbf{s}_j$ the per-cell profile score, are consistent for the asymptotic variance in (ii)—the latter remaining consistent under within-cell dependence misspecification with correctly specified conditional means; and (iv) $\hat{\gamma}_J$ attains the semiparametric efficiency bound for the model in which the α_j are unrestricted nuisance parameters, the conditional score being the efficient score.

Three comments. First, the mechanism is special to the Poisson–log pair: the profile score is *exactly* the conditional score, hence exactly unbiased for every J and m ; the incidental-parameter bias that plagues, e.g., the fixed-effects logit (where conditional ML exists but differs from profile ML, and profile ML has $O(1/m)$ bias (Andersen, 1970; Fernández-Val and Weidner, 2016)) is identically zero here. Section S4.4 exhibits the logit failure mode and its conditional-likelihood repair quantitatively, and Section S3.1 discusses how both the conditional likelihood (Chamberlain, 1980) and the analytic bias corrections of Hahn and Newey (2004) and Fernández-Val and Weidner (2016) slot into the cell-collapsed computational structure. This is the likelihood-theoretic content of the folklore that “Poisson FE is immune to the incidental-parameters problem” (Lancaster, 2000; Wooldridge, 1999), sharpened to a $J/N \rightarrow c$ limit theorem with variance estimation. Second, the theorem’s variance is the *conditional* information, which the naive full-information plug-in would overstate; the tilted-moment machinery produces the correct Schur complement at no extra cost. Third, event-free cells contribute no information and are dropped; this is the profile-likelihood counterpart of the separation drop rule and is accounted for in the theory by summing over $D_j > 0$.

Section S4.2 probes the theorem at $c = 1/2$ —one fixed effect per two observations—where nominal 95% intervals from both estimators in (iii) cover at 94–96%.

S3 Further extensions

S3.1 Families beyond Poisson

The framework of Section 2 of the main paper was stated for a generic exponential dispersion family; I record the two most used instances.

Logistic regression. With $b(\theta) = \log(1 + e^\theta)$, $\mu_i = (1 + e^{-\eta_i})^{-1}$ and $\omega_i = \mu_i(1 - \mu_i)$: the tilt weights each within-cell observation by its Bernoulli variance, so the tilted measure concentrates on observations near the decision boundary. All of Theorem 1 of the main paper, Corollary 1 of the main paper, and Theorems S1–S3 apply verbatim (the logit link is canonical). Unlike the Poisson case the tilt depends on $\boldsymbol{\delta}$ through ω_i , so the within-cell moments must be re-accumulated each iteration—which the single pass does anyway. The proportional-regime theory of Section S2.3 does *not* transfer: profiling and conditioning differ for the logit, and profile ML carries $O(1/m)$ incidental-parameter bias—for matched pairs ($m = 2$) the classical limit is $2\gamma_0$ (Andersen, 1970)—which Section S4.4 reproduces numerically. Two repairs are compatible with the cell-collapsed computational structure. *Conditional ML* (Chamberlain, 1980): conditioning on cell totals gives per-cell denominators that are elementary symmetric functions of $\{e^{\gamma' \mathbf{x}_{ij}}\}_{i \in j}$; these obey the standard Newton–Girard recursion and are computable in $O(m_j D_j)$ operations per cell—polynomial, not combinatorial—so for the bounded cell sizes of the proportional regime the conditional score and information are one more $O(N)$ cell pass (the reference implementation includes an exact conditional logit along these lines, used in Section S4.4). *Analytic bias correction* (Hahn and Newey, 2004; Fernández-Val and Weidner, 2016): the correction terms are averages of within-cell weighted moments of $(\mu_i, \omega_i, \partial \omega_i / \partial \eta)$ and leverage-type quantities, all of which are cell-decomposable in exactly the sense of Theorem 1 of the main paper and Proposition S2; there is no computational barrier to evaluating them inside the tilted pass, and I view a packaged implementation as worthwhile software work rather than an open methodological problem.

Gamma regression with log link. Here the log link is not canonical, Proposition 2 of the main paper applies, and the Fisher weights are identically one: the “tilted” moments are ordinary exposure-weighted cell moments, constant across iterations. One pass suffices for the whole optimization, and only the working residual sums R_j are refreshed—a family where the acceleration is at its most extreme.

S3.2 Penalized estimation

In applications with many continuous covariates (genomics: expression levels alongside batch cells), sparsity-inducing penalties on $\boldsymbol{\gamma}$ are natural while leaving $\boldsymbol{\delta}$ and the absorbed effects unpenalized. Consider

$$\min_{\boldsymbol{\gamma}, \boldsymbol{\delta}} -\ell(\boldsymbol{\gamma}, \boldsymbol{\delta}) + \lambda_1 \|\boldsymbol{\gamma}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\gamma}\|_2^2.$$

Proposition S1 (Penalized tilted-moment proximal Newton). *At each outer iteration, form the quadratic model of $-\ell$ from the tilted moments (S1) (one $O(Np^2)$ pass), i.e., the exact second-order expansion with gradient $-\mathbf{U}$ and curvature $\mathbf{F}$. (i) Ridge ($\lambda_1 = 0$): the update solves the $(p + k)$ -dimensional system with $\lambda_2 \mathbf{I}$ added to the $\boldsymbol{\gamma}$ -block; all conclusions of Corollary 1 of the main paper hold. (ii) Lasso/elastic net: one cycle of coordinate descent on the penalized*

quadratic costs $O((p+k)^2)$ —independent of N —after the pass, so the inner solver of [Friedman et al. \(2010\)](#) runs entirely on cell-collapsed quantities; the outer iteration is a proximal Newton method and converges quadratically to the penalized optimum near the solution under the conditions of [Lee et al. \(2014\)](#). (iii) The Karush–Kuhn–Tucker conditions, hence the active set and the regularization path, are functions of the tilted moments only.

The practical consequence: a lasso path for a Poisson model with $N = 10^7$, $J = 10^3$ cells, and $p = 50$ expression covariates costs per path point what a dense implementation spends per coordinate.

S3.3 Firth’s adjustment inside the tilted pass

Under separation (Section [S1.5](#)) the MLE does not exist; Firth’s bias-reducing penalty $\ell^*(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) + \frac{1}{2} \log \det \mathbf{F}(\boldsymbol{\theta})$ ([Firth, 1993](#)) guarantees a finite maximizer for binomial models ([Heinze and Schemper, 2002](#)) and removes the $O(N^{-1})$ bias term in general. For canonical links the adjusted score has the closed form $\mathbf{U}^* = \mathbf{U} + \sum_i h_i \frac{b'''(\eta_i)}{2b''(\eta_i)} \mathbf{a}_i$ with leverages $h_i = \omega_i \mathbf{a}_i' \mathbf{F}^{-1} \mathbf{a}_i$ ([Firth, 1993](#); [Kosmidis and Firth, 2009](#)).

Proposition S2 (Tilted Firth). *Given the partitioned inverse of $\mathbf{F}$ (available from the Newton solve), the leverages decompose as $h_i = \omega_i \{ \mathbf{x}_i' \mathbf{V}_{\gamma\gamma} \mathbf{x}_i + 2 \mathbf{x}_i' \mathbf{c}_{j(i)} + d_{j(i)} \}$ with cell-level precomputations $\mathbf{c}_j = \mathbf{V}_{\gamma\delta} \mathbf{w}_j$ and $d_j = \mathbf{w}_j' \mathbf{V}_{\delta\delta} \mathbf{w}_j$ at $O(J(kp + k^2))$; the adjusted score, and hence the entire Firth quasi-scoring iteration ([Kosmidis and Firth, 2009](#)), is computable at $O(N(p^2 + p)) + O(J(k^2 + kp)) + O((p + k)^3)$ per iteration—the same order as Corollary 1 of the main paper. For Poisson, $b'''/2b'' \equiv 1/2$; for logit it is $\frac{1}{2}(1 - 2\mu_i)$.*

Thus the statistically principled response to separation costs essentially nothing beyond the unpenalized fit, removing the last reason to respond to separation by ad hoc data deletion. Section [S4.3](#) quantifies the finite-sample behavior.

A clarification on software behavior is in order. Both implementations detect and drop separated *absorbed* groups automatically before estimation (Section [S1.5](#)). Neither switches to the Firth estimator automatically: separation induced by the β -block surfaces as flagged non-convergence (the Stata command sets `e(converged) = 0`; the reference implementation returns `converged=False`), and the Firth fit must be requested explicitly (`firth=True` in the reference implementation; the Stata command currently reports the diagnosis and does not expose a Firth option). The non-switching is deliberate: the Firth estimator is a different estimand-defining penalty, not a numerical fallback, and silently substituting it would change reported quantities without the user’s knowledge. A principled automatic *detector*—running the linear-programming certificate of [Correia et al. \(2019\)](#) on the cell aggregates, at $O(J)$ rather than $O(N)$ scale—is a natural software addition that the tilted representation makes cheap.

Table S1: Study 2: predicted versus empirical per-sweep contraction of weighted alternating projections; sweeps to $\|\Delta\|_\infty < 10^{-13}$.

ρ	Predicted $c_F^2 = \sigma_2^2$	Empirical rate	Sweeps
0.00	0.0024	0.0019	7
0.30	0.1897	0.1873	19
0.60	0.5567	0.5547	48
0.80	0.7542	0.7534	96
0.90	0.8896	0.8747	197
0.95	0.9469	0.9469	468
0.99	0.9911	0.9910	2487

S4 Simulation studies for the supplement results

The studies below validate the results of Sections S1–S3; they follow the computational protocol of main-paper Section 5 (single laptop core, fixed seeds, scripts `sim2–sim5` in the code supplement) and complete in minutes.

S4.1 Convergence rates of alternating projections

Study 2 validates Theorem S2(ii) and Corollary S1. Two factors ($J_1 = 100$, $J_2 = 80$, $N = 2 \times 10^5$) are assigned with a mixing parameter ρ : with probability ρ a unit’s level of factor 2 is deterministically tied to its level of factor 1, otherwise uniform—so ρ sweeps the weighted bipartite graph from expander-like ($\rho = 0$) to nearly disconnected ($\rho = 0.99$). At Poisson optimum weights I compute the predicted per-sweep rate σ_2^2 from the normalized biadjacency kernel, then measure the empirical contraction factor of weighted alternating-projections demeaning of a generic covariate.

Table S1 shows agreement to the third decimal across the entire range, including the near-disconnected regime where the rate is within 10^{-3} of 1 and the sweep count grows as $(1 - c_F^2)^{-1}$. The spectral prediction is computable before iterating (a sparse SVD of the 100×80 weighted cross-tabulation) and could be used to warn users, select orderings, or switch to conjugate-gradient acceleration when the gap is small (Gaure, 2013; Correia, 2016).

The second panel of the study probes Theorem S3 with $G = 3$ factors ($J_1, J_2, J_3 = 100, 80, 60$; factor 2 tied to factor 1 and factor 3 to factor 2, each with probability ρ). For each ρ I compute the exact level-space spectral rate of Theorem S3(ii) (function `gs_spectral_rate`, an eigencomputation on 240 levels), the Friedrichs product certificate of Theorem S3(iii), and the empirical per-sweep contraction of the three-factor demeaning.

Table S2 confirms the two halves of the theorem exactly as designed: the spectral rate matches the observed contraction to the third decimal at every ρ —the $G = 2$ precision carries over to $G = 3$ unchanged—while the a priori norm bound is valid everywhere but conservative, overstating the rate substantially in the well-conditioned regimes. The practical protocol this

Table S2: Study 2, $G = 3$ absorbed factors: exact spectral rate of Theorem S3(ii) versus empirical per-sweep contraction, with the product-of-sines norm certificate of Theorem S3(iii).

ρ	Exact ρ_{GS}	Empirical rate	Norm bound	Sweeps
0.00	0.0066	0.0064	0.0766	7
0.30	0.2520	0.2524	0.6218	23
0.60	0.5293	0.5317	0.8788	46
0.80	0.7907	0.7890	0.9763	113
0.90	0.8802	0.8837	0.9928	225
0.95	0.9548	0.9552	0.9989	569

Table S3: Study 3: profile (concentrated) Poisson estimation with $J/N = 1/m$; 1,000 replications; $\gamma_{01} = 0.3$. “Cover” is the empirical coverage of nominal 95% intervals from the profile information (P) and the cell-clustered sandwich (C).

N	m	J/N	Bias($\hat{\gamma}_1$)	SD($\hat{\gamma}_1$)	$\overline{\text{SE}}_P$	Cover _P	Cover _C
20,000	2	0.50	−0.0003	0.0156	0.0157	0.954	0.953
20,000	5	0.20	+0.0009	0.0118	0.0119	0.953	0.951
20,000	20	0.05	+0.0004	0.0107	0.0107	0.954	0.953
80,000	2	0.50	+0.0004	0.0080	0.0078	0.953	0.951
80,000	5	0.20	−0.0001	0.0059	0.0060	0.955	0.954
80,000	20	0.05	−0.0001	0.0054	0.0053	0.943	0.941

supports: use the product certificate (computable from angles factor-by-factor) for a quick feasibility screen, and the $O(\sum J_g)$ -dimensional spectral computation for the exact rate and for the identification diagnostic of Theorem S3(ii).

S4.2 The proportional regime

Study 3 probes Theorem S6 at its most extreme: Poisson cells of bounded size $m \in \{2, 5, 20\}$, one fixed effect per cell, heterogeneous cell effects $\alpha_j \sim \mathcal{N}(-1.2, 0.8^2)$, a covariate correlated with the cell effects (so that ignoring the effects biases $\hat{\gamma}$), $N \in \{20,000, 80,000\}$, 1,000 replications. With $m = 2$ the design estimates one nuisance parameter per two observations ($c = 1/2$).

Table S3 confirms every element of the theorem: bias is an order of magnitude below the Monte Carlo standard deviation even at $c = 1/2$ and shrinks with N at fixed c ; the profile-information standard errors track the true sampling variability to the third decimal; and both interval constructions cover at 94–96% throughout. For contrast, the same data analyzed with the cell effects ignored (a single intercept) biases the coefficient of the effect-correlated covariate by +0.28 (truth −0.2) and yields nominal 95% intervals with *zero* empirical coverage in every configuration—the point being that the fixed effects are not ignorable here, they are merely free.

Table S4: Study 4: quasi-separation frequency and estimator behavior; true $\gamma_z = 1.5$. MLE summaries condition on non-separated replications; Firth summaries use all replications. MAE is median absolute error.

N	Separated	Firth converged	$\text{med}(\hat{\gamma}_z^{\text{MLE}})$	$\text{med}(\hat{\gamma}_z^{\text{Firth}})$	MAE_{MLE}	$\text{MAE}_{\text{Firth}}$
300	0.394	1.000	1.704	1.586	0.587	0.498
600	0.042	1.000	1.581	1.556	0.446	0.400
1200	0.002	1.000	1.519	1.512	0.235	0.222
2400	0.000	1.000	1.520	1.515	0.199	0.195

Table S5: Study 5: incidental-parameter bias of the logit profile MLE at $J/N = 1/m$ versus the conditional MLE; true $\gamma_0 = 1$; 500 replications. The $m = 2$ profile mean approaches Andersen’s classical $2\gamma_0$ limit.

m	J/N	Mean $\hat{\gamma}^{\text{profile}}$	Bias ^{profile}	Mean $\hat{\gamma}^{\text{cond}}$	Bias ^{cond}
2	0.50	1.869	+0.869	1.000	−0.000
4	0.25	1.403	+0.403	0.998	−0.002
8	0.125	1.165	+0.165	0.998	−0.002

S4.3 Separation and the Firth-adjusted tilted pass

Study 4 stresses the boundary where Assumption S2 fails. Logistic model, $J = 10$ cells, a continuous covariate, and a rare binary exposure (3% prevalence, true log-odds 1.5): in small samples the exposed events can be quasi-separated with appreciable probability. I compare the MLE (tilted Newton, non-convergence or $|\hat{\gamma}_z| > 10$ flagged as separated) with the Firth-adjusted tilted iteration of Proposition S2, 500 replications per N .

At $N = 300$ the MLE fails to exist in 39% of replications; the Firth iteration converges in all 500, with smaller median bias *conditional on the MLE existing at all* and uniformly smaller median absolute error. As N grows the two estimators merge, as theory requires. The cost of the adjustment was a factor ~ 2 per iteration (the extra unadjusted pass for current-iterate leverages), consistent with Proposition S2.

S4.4 The logit failure mode and its conditional repair

Study 5 quantifies the boundary drawn in Sections S2.3 and S3.1: in the proportional regime the Poisson profile MLE is exactly unbiased (Theorem S6, Study 3), but the logit profile MLE is not. Cells of size $m \in \{2, 4, 8\}$, one logit fixed effect per cell, $\alpha_j \sim \mathcal{N}(0, 1)$, a single covariate with $\gamma_0 = 1$, $N = 8,000$, 500 replications; uninformative cells (all events or all non-events) are dropped, as both estimators require. I compare the profile MLE (inner concentration of the cell effects by vectorized Newton, outer Schur-complement steps—the logit analogue of `profile_poisson`) with the exact conditional MLE of Chamberlain (1980), computed cellwise as described in Section S3.1.

Table S5 displays the textbook pathology at full strength: the profile bias is 87% of the

parameter at $m = 2$ (approaching the $2\gamma_0$ limit of Andersen, 1970) and decays at the predicted $O(1/m)$, while the conditional estimator is unbiased to the third decimal at every m *at the same cell-collapsed computational cost*. Read together with Study 3, the two tables make the asymptotic message concrete: the proportional regime is not generically safe (logit) nor generically fatal (Poisson); it is the profile-conditional coincidence, an algebraic property of the family, that decides—and when the coincidence fails, the conditional likelihood is the drop-in repair within the same framework.

S-A Proofs for Section S1 of this supplement

Throughout this appendix write $D = q + \sum_g J_g$ for the total parameter dimension, $\mathbf{a}_i \in \mathbb{R}^D$ for the stacked design vector of observation i (individual-level, cell-level, and absorbed indicator parts), $\mathcal{K} = \{\mathbf{u} : \mathbf{a}'_i \mathbf{u} = 0 \ \forall i\}$ for the null space, and $E = \mathcal{K}^\perp \cong \mathbb{R}^D / \mathcal{K}$. By Assumption S1, $\ell(\boldsymbol{\theta}) = \phi^{-1} \sum_i \{y_i \mathbf{a}'_i \boldsymbol{\theta} - b(\mathbf{a}'_i \boldsymbol{\theta} + o_i)\}$ (absorbing offsets into b 's argument) is concave on $\mathbb{R}^D$, constant on cosets of $\mathcal{K}$, and its restriction $\bar{\ell}$ to E is *strictly* concave: for $\mathbf{u} \in E \setminus \{\mathbf{0}\}$, some $\mathbf{a}'_i \mathbf{u} \neq 0$ and $-\mathbf{u}' \nabla^2 \ell \mathbf{u} = \phi^{-1} \sum_i b''(\cdot)(\mathbf{a}'_i \mathbf{u})^2 > 0$.

Lemma S2 (Compact superlevel sets). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be strictly concave, continuous, and attain its maximum at x^* . Then every superlevel set $\{x : f(x) \geq a\}$, $a \leq f(x^*)$, is compact.*

Proof. Superlevel sets of concave functions are closed and convex. Suppose S_a is unbounded for some a ; then S_a contains a ray $\{x^* + t\mathbf{v} : t \geq 0\}$ for some unit vector $\mathbf{v}$ (a closed unbounded convex set containing x^* contains a ray from x^* ; see Rockafellar, 1970, Thm. 8.4). Let $\varphi(t) = f(x^* + t\mathbf{v})$: strictly concave on $[0, \infty)$, $\varphi(t) \geq a$ for all t , maximized at $t = 0$. If $\varphi'(t_0^+) < 0$ for some t_0 , concavity gives $\varphi(t) \leq \varphi(t_0) + \varphi'(t_0^+)(t - t_0) \rightarrow -\infty$, contradicting $\varphi \geq a$; hence the (strictly decreasing, by strict concavity) right derivative satisfies $\varphi'(t^+) \geq 0$ for all $t > 0$. But $\varphi'(0^+) \leq 0$ at the maximum, and strict decrease then forces $\varphi'(t^+) < 0$ for $t > 0$ —a contradiction. $\square$

S-A.1 Proof of Theorem S1

All statements are on E ; representatives are normalized after each sweep (any fixed measurable normalization, e.g., recentering each absorbed factor and compensating in the intercept, changes nothing in ℓ).

Monotonicity and confinement. Each $\boldsymbol{\alpha}^{(g)}$ -update is an exact maximization over its block (the block objective separates over the levels of factor g ; each level's scalar problem $\alpha \mapsto c_1 \alpha - \sum_i b(\alpha + \text{rest}_i)$ is strictly concave with derivative $\rightarrow \pm\infty$ prevented only if the level has an interior event, which Assumption S3(i) guarantees; hence a unique finite maximizer exists). The $\boldsymbol{\beta}$ -update is accepted only under the Armijo test, so $\ell(\boldsymbol{\theta}_{t+1}) \geq \ell(\boldsymbol{\theta}_t)$, and the sequence

$\ell(\boldsymbol{\theta}_t)$ converges to some $L^* \leq \ell(\boldsymbol{\theta}^*)$ (Assumption S2). By Lemma S2 applied to $\bar{\ell}$ on E , the normalized iterates remain in the compact set $\bar{S} = \{\bar{\ell} \geq \ell(\boldsymbol{\theta}_0)\}$.

Limit points are stationary. Let $\bar{\boldsymbol{\theta}}$ be a limit point of the normalized iterates, along t_r . The per-cycle increment $\ell(\boldsymbol{\theta}_{t+1}) - \ell(\boldsymbol{\theta}_t) \rightarrow 0$, and increments are sums of nonnegative per-block increments, so each block's increment vanishes. Fix a block g and let T_g denote the map that exactly maximizes over block g holding the rest fixed; T_g is continuous on $\bar{S}$ (the unique maximizer of a strictly concave, jointly continuous objective depends continuously on the fixed coordinates; Berge's maximum theorem plus uniqueness). If block g were not optimal at $\bar{\boldsymbol{\theta}}$, then $\varepsilon := \ell(T_g \bar{\boldsymbol{\theta}}) - \ell(\bar{\boldsymbol{\theta}}) > 0$; by continuity of T_g and ℓ , the block- g increment at $\boldsymbol{\theta}_{t_r}$ (with the preceding blocks of the cycle already updated, which also converge along the subsequence by the same argument applied inductively in the cycle order) exceeds $\varepsilon/2$ for large r —contradiction. Hence every absorbed block of $\bar{\boldsymbol{\theta}}$ is blockwise optimal, i.e., $\nabla_{\alpha} \ell(\bar{\boldsymbol{\theta}}) = \mathbf{0}$. For the β -block, suppose $\mathbf{U}_{\beta}(\bar{\boldsymbol{\theta}}) \neq \mathbf{0}$. On the compact set $\bar{S}$, the information $\mathbf{F}$ is continuous and, by Assumption S3(ii), $\mathbf{F}_{\beta\beta}$ is nonsingular with eigenvalues in a fixed interval $[\underline{\lambda}, \bar{\lambda}]$, $\underline{\lambda} > 0$; the Newton direction $\mathbf{d} = \mathbf{F}_{\beta\beta}^{-1} \mathbf{U}_{\beta}$ then satisfies $\mathbf{U}'_{\beta} \mathbf{d} \geq \|\mathbf{U}_{\beta}\|^2 / \bar{\lambda}$, and the standard backtracking bound guarantees an accepted step length λ_t bounded below and an increment $\geq \sigma \lambda_t \mathbf{U}'_{\beta} \mathbf{d}$ bounded away from zero in a neighborhood of $\bar{\boldsymbol{\theta}}$ (Bertsekas, 1999, Prop. 1.2.1 and Sec. 2.7)—again contradicting vanishing increments. Hence $\nabla \ell(\bar{\boldsymbol{\theta}}) = \mathbf{0}$.

Conclusion. For a concave differentiable function, a stationary point is a global maximizer, so $\bar{\boldsymbol{\theta}} = \boldsymbol{\theta}^*$ modulo $\mathcal{K}$ by strict concavity on E ; since every limit point of the normalized iterates in the compact set $\bar{S}$ equals the normalized $\boldsymbol{\theta}^*$, the whole sequence converges to it, and $L^* = \ell(\boldsymbol{\theta}^*)$, proving (i)–(ii). Part (iii) follows because the argument used only: exactness/uniqueness of the absorbed block maximizations, sufficient ascent of the β -step, and vanishing per-block increments—all invariant to replacing the β -step by exact maximization and to essentially-cyclic orderings (every block updated at least once every B iterations). In particular, running the inner α -loop to tolerance before each outer β -step (as Algorithm S1 does) is an instance of an essentially-cyclic ordering: each α -block is updated at least once, and the β -block exactly once, per outer cycle, so the vanishing-per-block-increment argument applies verbatim with B equal to the (finite, since the inner loop converges R -linearly by Theorem S2(i)) inner-loop length.

For part (iv), inspect where Assumption S1 entered: (a) concavity of ℓ on $\mathbb{R}^D$ and strict concavity on E used only that each per-observation contribution $\eta \mapsto \ell_i(\eta)$ is (strictly) concave, which is the log-concave-link hypothesis; (b) the absorbed block subproblems are separable sums of strictly concave scalar functions, so unique finite maximizers exist under the analogue of Assumption S3(i) (for the gamma-log family every group maximizer is finite whenever the group is nonempty, since $y_i > 0$ a.s.); and (c) the sufficient-ascent argument for the β -step used only that the step direction is $\mathbf{d} = \mathbf{P}^{-1} \mathbf{U}_{\beta}$ for a symmetric preconditioner $\mathbf{P}$ with eigenvalues in $[\underline{\lambda}, \bar{\lambda}]$, giving $\mathbf{U}'_{\beta} \mathbf{d} \geq \|\mathbf{U}_{\beta}\|^2 / \bar{\lambda}$ —satisfied by the expected information of Fisher scoring under Assumption S3(ii). $\square$

S-A.2 Proof of Theorem S2

(i) Write the inner objective as $f(\boldsymbol{\alpha}) = \ell(\boldsymbol{\beta}, \boldsymbol{\alpha})$ with fixed $\boldsymbol{\beta}$, and let $\hat{\boldsymbol{\alpha}}$ be its maximizer (unique after normalization). The block- g exact update solves $\nabla_g f(\boldsymbol{\alpha}^{(<g),+}, \boldsymbol{\alpha}^{(g)}, \boldsymbol{\alpha}^{(>g)}) = \mathbf{0}$; the block Hessian $\mathbf{F}_{gg}$ is diagonal with entries $\sum_{i \in \text{level}} b''(\eta_i) > 0$, so by the implicit function theorem (using $b \in C^3$) the update map is C^1 near the fixed point with partial derivatives $-\mathbf{F}_{gg}^{-1} \mathbf{F}_{g,h}$ with respect to block $h \neq g$. Composing the G block updates in cycle order gives $DT(\hat{\boldsymbol{\alpha}}) = -(\mathbf{D} + \mathbf{L})^{-1} \mathbf{L}'$, where $\mathbf{F}_{\alpha\alpha} = \mathbf{D} + \mathbf{L} + \mathbf{L}'$ is the block diagonal/lower/upper decomposition—the classical Gauss–Seidel error operator—restricted to the quotient. Since $\mathbf{F}_{\alpha\alpha}$ is symmetric positive definite on the quotient, $\rho_{\text{GS}} = \rho(DT) < 1$ (Golub and Van Loan, 2013, Sec. 11.2), and Ostrowski’s theorem (Ortega and Rheinboldt, 1970, Sec. 10.1) yields local R -linear convergence at rate ρ_{GS} .

(ii) Let $G = 2$ with indicator matrices $\mathbf{A}_1 \in \mathbb{R}^{N \times J_1}$, $\mathbf{A}_2 \in \mathbb{R}^{N \times J_2}$ and $\mathbf{F}_{gh} = \mathbf{A}'_g \Omega^* \mathbf{A}_h$. The eigenvalues of the two-block Gauss–Seidel operator are $\{0\}$ together with the eigenvalues of $\mathbf{F}_{22}^{-1} \mathbf{F}_{21} \mathbf{F}_{11}^{-1} \mathbf{F}_{12}$ (eliminate the first block). In the Ω^* inner product this matrix represents $\mathbf{P}_2 \mathbf{P}_1|_{V_2}$, the compression to V_2 of the product of the orthogonal projections onto V_1 and V_2 ; its spectrum on the complement of $V_1 \cap V_2$ is $\{\cos^2 \theta_i\}$ with θ_i the principal angles between $V_1 \ominus (V_1 \cap V_2)$ and $V_2 \ominus (V_1 \cap V_2)$, whose supremum-cosine is c_F (Deutsch, 2001, Ch. 9). On the quotient (which removes $V_1 \cap V_2$, the flat directions), $\rho = c_F^2$. The identical rate for the demeaning residuals is the theorem of Aronszajn (1950) and Kayalar and Weinert (1988): $\|(\mathbf{P}_{V_2^\perp} \mathbf{P}_{V_1^\perp})^n - \mathbf{P}_{(V_1+V_2)^\perp}\|_{\Omega^*} = c_F^{2n-1}$, using that the Friedrichs angle is invariant under orthogonal complementation (Deutsch, 2001, Ch. 9).

(iii) Append the $\boldsymbol{\beta}$ -block: the cycle map with exact maximizations is again C^1 by the implicit function theorem (the $\boldsymbol{\beta}$ -block Hessian is nonsingular by Assumption S3(ii)), its Jacobian is the block Gauss–Seidel operator of the full $\mathbf{F}(\boldsymbol{\theta}^*)$ in the stated ordering, and the SPD argument of (i) applies on the quotient. $\square$

S-A.3 Proof of Corollary S1

Represent $u \in V_1$, $v \in V_2$ by their level vectors, $u = \mathbf{A}_1 \tilde{u}$, $v = \mathbf{A}_2 \tilde{v}$. Then $\langle u, v \rangle_{\Omega^*} = \tilde{u}' \mathbf{B} \tilde{v}$, $\|u\|_{\Omega^*}^2 = \tilde{u}' \mathbf{D}_1 \tilde{u}$, $\|v\|_{\Omega^*}^2 = \tilde{v}' \mathbf{D}_2 \tilde{v}$. Substituting $\mathbf{r} = \mathbf{D}_1^{1/2} \tilde{u}$, $\mathbf{s} = \mathbf{D}_2^{1/2} \tilde{v}$ turns the Friedrichs supremum into the second singular value of $\mathbf{T} = \mathbf{D}_1^{-1/2} \mathbf{B} \mathbf{D}_2^{-1/2}$: the top singular pair $(\mathbf{D}_1^{1/2} \mathbf{1}, \mathbf{D}_2^{1/2} \mathbf{1})$, with $\sigma_1 = 1$, spans exactly the image of the intersection $V_1 \cap V_2 \supseteq \text{constants}$, which the Friedrichs angle excludes. $\mathbf{T}$ is the transition kernel of the two-step random walk on the weighted bipartite graph with biadjacency $\mathbf{B}$; by Perron–Frobenius, the multiplicity of $\sigma = 1$ equals the number of connected components, so $\sigma_2 < 1$ iff the graph is connected, which is also exactly the condition $V_1 \cap V_2 = \text{span}\{\mathbf{1}\}$ (Abowd et al., 1999; Gaure, 2013). $\square$

S-A.4 Proof of Theorem S3

(i) Within one factor the indicator columns have disjoint supports, so $\mathbf{A}'_g \Omega^* \mathbf{A}_g = \mathbf{D}_g$ (diagonal); for $g \neq h$, $\mathbf{A}'_g \Omega^* \mathbf{A}_h = \mathbf{B}_{gh}$ by definition. Theorem S2(i) identifies the local rate with ρ of the block Gauss–Seidel operator of $\mathcal{F}$, which is a function of these blocks only.

(ii) *Eigenspace of 1.* $\mathbf{M}v = v \iff -\mathcal{L}'v = (\mathcal{D} + \mathcal{L})v \iff \mathcal{F}v = \mathbf{0}$, so the fixed points of the sweep are exactly the null vectors of $\mathcal{F}$; for connected designs these are the $G - 1$ between-factor shift directions, and in general the flats of the identification quotient. *Semisimplicity.* Suppose the eigenvalue 1 had a nontrivial Jordan block: then there is v with $(\mathbf{M} - \mathbf{I})v = f$ for some $f \in \text{null}(\mathcal{F}) \setminus \{\mathbf{0}\}$. Since $\mathbf{M} - \mathbf{I} = -(\mathcal{D} + \mathcal{L})^{-1}\mathcal{F}$, this reads $\mathcal{F}v = -(\mathcal{D} + \mathcal{L})f$. Multiply on the left by f' : the left side is $f'\mathcal{F}v = (\mathcal{F}f)'v = 0$. For the right side, $\mathcal{F}f = \mathbf{0}$ gives $(\mathcal{D} + \mathcal{L})f = -\mathcal{L}'f$, hence $f'\mathcal{D}f + 2f'\mathcal{L}f = f'\mathcal{F}f = 0$, i.e. $f'\mathcal{L}f = -\frac{1}{2}f'\mathcal{D}f$, and therefore $f'(\mathcal{D} + \mathcal{L})f = \frac{1}{2}f'\mathcal{D}f > 0$ because $\mathcal{D}$ is positive definite (every level carries positive weight) and $f \neq \mathbf{0}$. The two sides disagree—contradiction; so the eigenvalue 1 is semisimple and $\mathbb{R}\sum J_g = \text{null}(\mathcal{F}) \oplus W$ with W an $\mathbf{M}$ -invariant complement on which $\rho(\mathbf{M}|_W) = \rho_{\text{GS}} < 1$ by Theorem S2(i) (this is the semiconvergence of Gauss–Seidel for consistent singular symmetric positive semidefinite systems; cf. Keller, 1965; Berman and Plemmons, 1994). *Differences.* Decomposing an arbitrary start $e_0 = f_0 + w_0$, the iterates are $e_t = f_0 + \mathbf{M}^t w_0$, so $e_{t+1} - e_t = (\mathbf{M} - \mathbf{I})\mathbf{M}^t w_0$, whose norm ratios converge to $\rho(\mathbf{M}|_W)$ for generic w_0 : the fixed-point component drops out automatically, which is what makes plain power iteration on differences a valid estimator of ρ_{GS} . *Identification.* $\rho_{\text{GS}} < 1$ on the quotient iff $\mathcal{F}$ is positive definite there (Gauss–Seidel on a symmetric system converges iff the matrix is positive definite on the relevant space; Golub and Van Loan, 2013, Sec. 11.2), which is the definition of the multi-way design being identified up to the flats. *Cost.* One application of $\mathbf{M}$ is one block forward-substitution: for each g in order, a diagonal solve after accumulating $\sum_{h \neq g} \mathbf{B}_{gh} v_h$, i.e., $O(\sum_{g < h} \text{nnz}(\mathbf{B}_{gh}))$ operations.

(iii) Work in the Ω^* inner product on $\mathbb{R}^N$ and let $\varepsilon_t \in V_1 + \dots + V_G$ be the error of the fitted fixed-effect vector after sweep t . Updating block g to its exact blockwise maximizer replaces ε by $(\mathbf{I} - \mathbf{P}_g)\varepsilon$, where $\mathbf{P}_g$ is the Ω^* -orthogonal projection onto V_g : the block update minimizes the quadratic (IRLS) energy over corrections in V_g , and the minimizer of $\|\varepsilon + \delta\|_{\Omega^*}$ over $\delta \in V_g$ is $\delta = -\mathbf{P}_g \varepsilon$. One sweep therefore applies $\mathbf{E} = (\mathbf{I} - \mathbf{P}_G) \dots (\mathbf{I} - \mathbf{P}_1)$ —the multiplicative subspace-correction (equivalently, cyclic alternating projections onto the complements $M_g = V_g^\perp$)—and the level-space operator $\mathbf{M}$ of (ii) is the matrix of $\mathbf{E}|_{V_1 + \dots + V_G}$ in the (redundant) level parametrization, so the two have the same nonzero spectrum and $\rho_{\text{GS}} = \rho(\mathbf{E}|_{(\cap_g M_g)^\perp \cap (V_1 + \dots + V_G)}) \leq \|\mathbf{E} - \mathbf{P}_{\cap_g M_g}\|_{\Omega^*}$. The product bound of Smith et al. (1977) (see also Deutsch and Hundal, 1997; Badea et al., 2012) gives $\|\mathbf{E} - \mathbf{P}_{\cap_g M_g}\|_{\Omega^*} \leq (1 - \prod_{g=1}^{G-1} (1 - \tilde{c}_g^2))^{1/2}$ with $\tilde{c}_g = c(M_g, M_{g+1} \cap \dots \cap M_G)$; by the complement duality of the Friedrichs angle, $c(A, B) = c(A^\perp, B^\perp)$ (Deutsch, 2001, Ch. 9), and $M_{g+1} \cap \dots \cap M_G = (V_{g+1} + \dots + V_G)^\perp$, so $\tilde{c}_g = c_g$ as defined in the statement. Each c_g is the largest cosine between $V_g \ominus$ (intersection) and the

pooled span of the later factors, i.e., the largest non-unit generalized singular value of the level-space pencil formed from $\mathbf{D}_g$, the stacked cross-tabulations $(\mathbf{B}_{g,g+1}, \dots, \mathbf{B}_{g,G})$, and the pooled Gram matrix of the later factors—computable by a small dense or sparse eigenproblem. Positivity of all angles ($c_g < 1$) is equivalent to $V_g \cap (V_{g+1} + \dots + V_G) = \text{flats}$ for each g , which is identification.

(iv) For $G = 2$ the sweep operator eigenvalues are those of $\mathbf{D}_2^{-1} \mathbf{B}_{21} \mathbf{D}_1^{-1} \mathbf{B}_{12}$ (Appendix S-A, proof of Theorem S2(ii)), whose non-unit spectrum is $\{\sigma_i^2\}$ of the normalized biadjacency kernel, giving $\rho_{\text{GS}} = \sigma_2^2 = c_F^2$; the bound in (iii) has a single factor, $(1 - (1 - c_F^2))^{1/2} = c_F$. $\square$

S-B Proofs for Section S2 of this supplement

S-B.1 Proof of Theorem S4

Under Assumption S4, the divergence and normalization conditions of Fahrmeir and Kaufmann (1985, Sec. 3) hold for the canonical-link exponential family with bounded covariates—namely $\lambda_{\min}(\mathbf{F}_N) \rightarrow \infty$ and $\mathbf{F}_N^{-1/2} \mathbf{F}_N(\boldsymbol{\theta}) \mathbf{F}_N^{-1/2} \rightarrow \mathbf{I}$ uniformly on shrinking $\mathbf{F}_N$ -neighborhoods of $\boldsymbol{\theta}_0$, which follow from Assumption S4(i)–(ii) (bounded covariates with $b \in C^3$ give the uniform third-derivative control, and $N^{-1} \mathbf{F}_N \rightarrow \mathbf{F}_\infty \succ \mathbf{0}$ gives the normalization). Their normed information $\mathbf{F}_N^{-1/2}$ exists for large N , and their Theorems 1–2 give existence, consistency, and $\mathbf{F}_N^{1/2}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \phi \mathbf{I})$; Assumption S4(ii) converts this to the stated $\sqrt{N}$ form. Efficiency is the standard exponential-family Cramér–Rao attainment. Under correct mean but misspecified distribution, the estimator is the PML estimator of Gouriéroux et al. (1984) for the linear exponential family, whose Theorem 3 gives the sandwich limit; consistency of the cell-accumulated sandwich estimator follows from the law of large numbers applied to $\hat{u}_i^2$ -weighted design moments, using the uniform consistency of $\hat{\mu}_i$ on bounded designs. $\square$

S-B.2 Proof of Theorem S5

Normalize $\phi = 1$. Write $\mathbf{F} = \mathbf{F}_N(\boldsymbol{\theta}_0)$ (positive definite for large N by Assumption S5(ii), which presupposes invertibility), $\mathbf{U} = \mathbf{U}(\boldsymbol{\theta}_0)$, and change variables $\mathbf{s} = \mathbf{F}^{1/2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)$, $g(\mathbf{s}) = \ell(\boldsymbol{\theta}_0 + \mathbf{F}^{-1/2} \mathbf{s})$. Then g is concave, $\nabla g(\mathbf{0}) = \mathbf{F}^{-1/2} \mathbf{U}$, and $\nabla^2 g(\mathbf{s}) = -\mathbf{F}^{-1/2} \mathbf{H}(\boldsymbol{\theta}_s) \mathbf{F}^{-1/2}$.

Step 1: score bound. $\mathbb{E} \|\nabla g(\mathbf{0})\|^2 = \text{tr}(\mathbf{F}^{-1/2} \text{Var}(\mathbf{U}) \mathbf{F}^{-1/2}) = q_N$ by the information equality, so $\|\nabla g(\mathbf{0})\| = O_p(\sqrt{q_N})$.

Step 2: uniform curvature control. Fix $r \geq 1$ and let $\|\mathbf{s}\| \leq r\sqrt{q_N}$. By Assumption S5(i,iv), $\omega_i = b''(\eta_{0i}) \in [\underline{\omega}, \bar{\omega}]$ with $0 < \underline{\omega} \leq \bar{\omega} < \infty$, and the balanced-leverage condition gives $\|\mathbf{F}^{-1/2} \mathbf{a}_i\|^2 = h_i/\omega_i \leq C q_N/(N \underline{\omega})$. Hence

$$\max_i |\eta_i(\boldsymbol{\theta}_s) - \eta_{0i}| = \max_i |\mathbf{a}_i' \mathbf{F}^{-1/2} \mathbf{s}| \leq r\sqrt{q_N} \cdot \sqrt{C q_N/(N \underline{\omega})} = C_r q_N/\sqrt{N} =: \varepsilon_N \rightarrow 0$$

using $q_N^2/N \rightarrow 0$. With L the Lipschitz constant of b'' on the relevant compact set, $|\sum_i \{b''(\eta_i(\boldsymbol{\theta}_s)) - b''(\eta_{0i})\}(\mathbf{a}_i' \mathbf{u})^2| \leq (L\varepsilon_N/\underline{\omega}) \mathbf{u}' \mathbf{F} \mathbf{u}$ for all $\mathbf{u}$, so uniformly on the ball,

$$-(1 + \delta_N) \mathbf{I} \preceq \nabla^2 g(\mathbf{s}) \preceq -(1 - \delta_N) \mathbf{I}, \quad \delta_N := L\varepsilon_N/\underline{\omega} \rightarrow 0. \quad (\text{S4})$$

Step 3: existence and rate. On $\|\mathbf{s}\| = r\sqrt{q_N}$, a second-order Taylor expansion with (S4) gives $g(\mathbf{s}) - g(\mathbf{0}) \leq \|\nabla g(\mathbf{0})\| r\sqrt{q_N} - \frac{1}{2}(1 - \delta_N)r^2 q_N$. On the event $\{\|\nabla g(\mathbf{0})\| \leq K\sqrt{q_N}\}$, whose probability exceeds $1 - K^{-2}\mathbb{E}\|\nabla g(\mathbf{0})\|^2/q_N = 1 - K^{-2}$, the right side is negative for $r > 4K$; concavity of g then confines the (unique, by strict concavity) maximizer $\hat{\mathbf{s}}$ to the open ball, proving existence and $\|\hat{\mathbf{s}}\| = \|\mathbf{F}^{1/2}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\| = O_p(\sqrt{q_N})$.

Step 4: asymptotic linearity. By the integral form of the mean value theorem, $\mathbf{0} = \nabla g(\hat{\mathbf{s}}) = \nabla g(\mathbf{0}) - \tilde{\mathbf{J}}\hat{\mathbf{s}}$ with $\tilde{\mathbf{J}} = -\int_0^1 \nabla^2 g(t\hat{\mathbf{s}}) dt$; by (S4), $\|\tilde{\mathbf{J}} - \mathbf{I}\| \leq \delta_N$, so

$$\|\hat{\mathbf{s}} - \nabla g(\mathbf{0})\| \leq \frac{\delta_N}{1 - \delta_N} \|\nabla g(\mathbf{0})\| = O_p\left(\frac{q_N}{\sqrt{q_N}}\sqrt{q_N}\right) = O_p(q_N^{3/2}/\sqrt{N}) = o_p(1)$$

under $q_N^3/N \rightarrow 0$.

Step 5: contrasts. For $\mathbf{C}_N$ with $\boldsymbol{\Sigma}_{C,N} = \mathbf{C}_N' \mathbf{F}^{-1} \mathbf{C}_N \rightarrow \boldsymbol{\Sigma}_C \succ \mathbf{0}$ and any fixed $\mathbf{v} \in \mathbb{R}^d$, write $Z_N = \mathbf{v}' \boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N'(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \mathbf{v}' \boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N' \mathbf{F}^{-1/2} \hat{\mathbf{s}}$. The remainder from Step 4 contributes $\leq \|\boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N' \mathbf{F}^{-1/2}\| o_p(1) = o_p(1)$ (the matrix has orthonormal-bounded rows by construction). The main term is $\sum_i c_{N,i}(y_i - b'(\eta_{0i}))$ with $c_{N,i} = \mathbf{v}' \boldsymbol{\Sigma}_{C,N}^{-1/2} \mathbf{C}_N' \mathbf{F}^{-1} \mathbf{a}_i$: independent, mean-zero, total variance $\mathbf{v}' \mathbf{v}(1 + o(1))$, and $\max_i |c_{N,i}| \leq \|\mathbf{v}\| \|\boldsymbol{\Sigma}_{C,N}^{-1/2}\| \|\mathbf{C}_N' \mathbf{F}^{-1/2}\| \max_i \|\mathbf{F}^{-1/2} \mathbf{a}_i\| = O(\sqrt{q_N/N}) \rightarrow 0$; since the residuals have uniformly bounded $(2 + \epsilon)$ moments on compact η -sets, the Lindeberg condition holds and $Z_N \xrightarrow{d} \mathcal{N}(0, \|\mathbf{v}\|^2)$; Cramér–Wold completes the claim. Consistency of the plug-in variance follows from (S4) evaluated at $\hat{\mathbf{s}}$: $\|\mathbf{F}^{-1/2} \mathbf{F}(\hat{\boldsymbol{\theta}}) \mathbf{F}^{-1/2} - \mathbf{I}\| \leq \delta_N' \xrightarrow{p} 0$, which implies relative consistency of $\mathbf{C}_N' \mathbf{F}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{C}_N$. $\square$

S-B.3 Proof of Lemma S1

For fixed $\boldsymbol{\gamma}$ the inner maximization separates over cells; $\partial \ell / \partial \alpha_j = D_j - e^{\alpha_j} E_j(\boldsymbol{\gamma})$ with $E_j(\boldsymbol{\gamma}) = \sum_i e^{\boldsymbol{\gamma}' \mathbf{x}_{ij} + o_{ij}}$, giving $e^{\hat{\alpha}_j} = D_j / E_j$ for $D_j > 0$ (for $D_j = 0$ the supremum is approached as $\alpha_j \rightarrow -\infty$ and the cell contributes a $\boldsymbol{\gamma}$ -free constant). Substituting,

$$\ell_p(\boldsymbol{\gamma}) = \sum_{j: D_j > 0} \left\{ \sum_i y_{ij}(\boldsymbol{\gamma}' \mathbf{x}_{ij} + o_{ij}) - D_j \log E_j(\boldsymbol{\gamma}) \right\} + \sum_j D_j (\log D_j - 1).$$

The bracket equals $\sum_i y_{ij} \log \pi_{ij}(\boldsymbol{\gamma})$ plus terms free of $\boldsymbol{\gamma}$, which is the multinomial log likelihood given D_j up to the combinatorial constant; and the multinomial law given the total is the standard Poisson-splitting identity. Concavity: $\boldsymbol{\gamma} \mapsto \log E_j(\boldsymbol{\gamma})$ is log-sum-exp of affine maps, hence convex. Differentiating the bracket gives the stated score and, again, negative Hessian $\sum_j D_j \text{Var}_{\pi_j}(\mathbf{x}_j)$; the Schur form follows by evaluating (S1) at $\hat{\boldsymbol{\alpha}}(\boldsymbol{\gamma})$, where $\mu_{ij} = D_j \pi_{ij}$, so

$M_j^{(0)} = D_j$, $\mathbf{m}_j = D_j \mathbb{E}_{\pi_j}[\mathbf{x}_j]$, and $\mathbf{Q} - \sum_j \mathbf{m}_j \mathbf{m}_j' / M_j^{(0)} = \sum_j D_j \{\mathbb{E}_{\pi_j}[\mathbf{x} \mathbf{x}'] - \mathbb{E}_{\pi_j}[\mathbf{x}] \mathbb{E}_{\pi_j}[\mathbf{x}']\}$. $\square$

S-B.4 Proof of Theorem S6

Throughout, expectations condition on the (fixed) covariates, offsets, and cell effects; cells are independent. Let $\ell_j(\gamma)$ denote cell j 's conditional (multinomial) log likelihood and $L_J(\gamma) = J^{-1} \sum_j \{\ell_j(\gamma) - \ell_j(\gamma_0)\}$, so that $\hat{\gamma}_J$ maximizes the concave function L_J by Lemma S1.

(i) *Consistency.* Each summand satisfies, on the compact set Γ , $|\ell_j(\gamma) - \ell_j(\gamma_0)| \leq 2D_j \sup_{i, \gamma \in \Gamma} |(\gamma - \gamma_0)' \mathbf{x}_{ij}| \leq C_\Gamma D_j$, and by Assumption S6(ii) the cell means are bounded, so D_j has uniformly bounded moments of all orders; hence $\text{Var}\{L_J(\gamma)\} \leq C/J \rightarrow 0$ and $L_J(\gamma) - \mathbb{E}L_J(\gamma) \xrightarrow{p} 0$ pointwise. The mean is a per-cell Kullback–Leibler divergence: $\mathbb{E}\{\ell_j(\gamma) - \ell_j(\gamma_0)\} = -\mathbb{E} \text{KL}(\pi_j(\gamma_0) \parallel \pi_j(\gamma)) \leq 0$, and a second-order expansion at γ_0 with Assumption S6(iii) gives an $\epsilon_0 > 0$ and, for large J , $\mathbb{E}L_J(\gamma) \leq -\frac{\lambda}{4} \|\gamma - \gamma_0\|^2$ whenever $\|\gamma - \gamma_0\| \leq \epsilon_0$; concavity of $\mathbb{E}L_J$ (with $\mathbb{E}L_J(\gamma_0) = 0$) extends the strict separation to all of Γ : for $\|\gamma - \gamma_0\| \geq \epsilon$ with $\epsilon \leq \epsilon_0$, evaluating along the segment from γ_0 yields $\mathbb{E}L_J(\gamma) \leq -\frac{\lambda \epsilon^2}{4}$. By the convexity lemma (Andersen and Gill, 1982; Pollard, 1991; Niemiro, 1992), pointwise convergence of the concave functions $L_J - \mathbb{E}L_J$ to 0 is automatically uniform on compacts, and the argmax of L_J converges in probability to γ_0 .

(ii) *Normality.* The profile score at the truth is $\mathbf{U}_J(\gamma_0) = \sum_j \mathbf{s}_j$ with $\mathbf{s}_j = \sum_i (y_{ij} - D_j \pi_{ij}(\gamma_0)) \mathbf{x}_{ij}$. Conditionally on D_j the multinomial model is correctly specified, so $\mathbb{E}[\mathbf{s}_j \mid D_j] = \mathbf{0}$, hence $\mathbb{E} \mathbf{s}_j = \mathbf{0}$ exactly, for every J and every configuration of cell effects—this is the absence of incidental-parameter bias. The summands are independent with $\text{Var}(\mathbf{s}_j) = \mathbb{E}[D_j \text{Var}_{\pi_j}(\mathbf{x}_j)]$ (conditional information equality), so $\text{Var}(J^{-1/2} \mathbf{U}_J) = \bar{\mathbf{F}}_J(\gamma_0)$, and $\|\mathbf{s}_j\| \leq 2D_j \max_i \|\mathbf{x}_{ij}\|$ has bounded third moments, so Lyapunov's CLT gives $J^{-1/2} \mathbf{U}_J \xrightarrow{d} \mathcal{N}(\mathbf{0}, \bar{\mathbf{F}})$ by Assumption S6(iii). For the Hessian, $J^{-1} \mathbf{H}_p(\gamma) = J^{-1} \sum_j D_j \text{Var}_{\pi_j(\gamma)}(\mathbf{x}_j)$ is an average of independent terms, bounded by CD_j and Lipschitz in γ with constant $\leq C'D_j$ on Γ ; a standard bracketing/finite-cover argument gives $\sup_{\|\gamma - \gamma_0\| \leq \epsilon_J} \|J^{-1} \mathbf{H}_p(\gamma) - \bar{\mathbf{F}}_J(\gamma_0)\| \xrightarrow{p} 0$ for any $\epsilon_J \rightarrow 0$. A Taylor expansion of the score around γ_0 , solved at $\hat{\gamma}_J$, then yields $\sqrt{J}(\hat{\gamma}_J - \gamma_0) = \bar{\mathbf{F}}_J^{-1} J^{-1/2} \mathbf{U}_J + o_p(1)$ and the stated limit.

(iii) *Variance estimation.* Consistency of $J^{-1} \mathbf{H}_p(\hat{\gamma}_J)$ for $\lim \bar{\mathbf{F}}_J$ is the uniform convergence just shown plus (i). For the clustered estimator, $J^{-1} \sum_j \mathbf{s}_j(\hat{\gamma}) \mathbf{s}_j(\hat{\gamma})'$: at γ_0 the LLN with bounded fourth moments gives convergence to $\lim J^{-1} \sum_j \text{Var}(\mathbf{s}_j)$; the difference between evaluation at $\hat{\gamma}$ and at γ_0 is bounded by a Lipschitz-in- γ envelope with integrable constant times $\|\hat{\gamma} - \gamma_0\| \xrightarrow{p} 0$. If the within-cell joint distribution is misspecified but the conditional mean $\mathbb{E}[y_{ij} \mid D_j] = D_j \pi_{ij}(\gamma_0)$ is correct, then $\mathbb{E} \mathbf{s}_j = \mathbf{0}$ still holds, the same argument applies with $\text{Var}(\mathbf{s}_j)$ no longer equal to the conditional information, and the sandwich $\mathbf{H}_p^{-1}(\sum_j \mathbf{s}_j \mathbf{s}_j') \mathbf{H}_p^{-1}$ remains consistent; the $G/(G-1)$ correction is asymptotically negligible.

(iv) *Efficiency.* In the model with unrestricted $(\alpha_1, \alpha_2, \dots)$, the nuisance tangent space at (γ_0, α_0) is spanned, cell by cell, by the α_j -scores $u_j = \sum_i (y_{ij} - \mu_{ij})$. The efficient score for γ is

the residual of the projection of the γ -score onto this space; by independence across cells the projection is cellwise, and within cell j , $\mathbb{E}[(\sum_i (y_{ij} - \mu_{ij}) \mathbf{x}_{ij}) u_j] = \sum_i \mu_{ij} \mathbf{x}_{ij}$, $\mathbb{E}[u_j^2] = \sum_i \mu_{ij}$, so the projection residual is $\sum_i (y_{ij} - \mu_{ij}) \mathbf{x}_{ij} - \frac{\sum_i \mu_{ij} \mathbf{x}_{ij}}{\sum_i \mu_{ij}} \sum_i (y_{ij} - \mu_{ij}) = \mathbf{s}_j$ evaluated at the truth: exactly the conditional score. The asymptotic variance in (ii) equals the inverse of the efficient information, so the bound is attained (van der Vaart, 1998, Ch. 25). $\square$

S-C Proofs for Section S3 of this supplement

S-C.1 Proof of Proposition S1

(i) Adding $\frac{\lambda_2}{2} \|\gamma\|^2$ contributes $\lambda_2 \mathbf{I}_p$ to the γ -block of the curvature and $-\lambda_2 \gamma$ to its gradient; the modified system has the same dimensions and cost. (ii) The proximal Newton step minimizes the penalized quadratic model $m(\boldsymbol{\theta}) = -\mathbf{U}'(\boldsymbol{\theta} - \boldsymbol{\theta}_t) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_t)' \mathbf{F}(\boldsymbol{\theta} - \boldsymbol{\theta}_t) + \lambda_1 \|\gamma\|_1$ (plus the ridge term); one coordinate update needs the corresponding row of $\mathbf{F}$ and the current residual of the linear system, both dense in $\mathbb{R}^{p+k}$, so a full cycle costs $O((p+k)^2)$ —all references to data enter through $(\mathbf{U}, \mathbf{F})$, which Theorem 1 of the main paper provides from one pass. Since $-\ell$ is convex with locally Lipschitz Hessian and $\mathbf{F} \succ \mathbf{0}$ near the solution, the inexact proximal Newton method converges locally quadratically (Lee et al., 2014, Thms. 3.4–3.5). (iii) The KKT conditions are $|\mathbf{U}_r - [\mathbf{F}(\boldsymbol{\theta} - \boldsymbol{\theta}_t)]_r| \leq \lambda_1$ for inactive γ_r , with equality sign structure for active ones: functions of $(\mathbf{U}, \mathbf{F})$ only. $\square$

S-C.2 Proof of Proposition S2

For canonical links, $\mathbf{F}(\boldsymbol{\theta}) = \sum_i b''(\eta_i) \mathbf{a}_i \mathbf{a}_i'$ depends on $\boldsymbol{\theta}$ through the η_i , and $\partial \mathbf{F} / \partial \theta_r = \sum_i b'''(\eta_i) a_{ir} \mathbf{a}_i \mathbf{a}_i'$. Firth's adjustment (Firth, 1993) is $A_r = \frac{1}{2} \text{tr}(\mathbf{F}^{-1} \partial \mathbf{F} / \partial \theta_r) = \frac{1}{2} \sum_i b'''(\eta_i) a_{ir} \mathbf{a}_i' \mathbf{F}^{-1} \mathbf{a}_i = \sum_i h_i \frac{b'''(\eta_i)}{2b''(\eta_i)} a_{ir}$ with $h_i = b''(\eta_i) \mathbf{a}_i' \mathbf{F}^{-1} \mathbf{a}_i$: the adjusted score is the ordinary score with working residual $y_i - b'(\eta_i) + h_i b'''(\eta_i) / 2b''(\eta_i)$, which the cell accumulation of Theorem 1 of the main paper ingests unchanged. For the leverages, write the partitioned inverse blocks $\mathbf{V}_{\gamma\gamma}, \mathbf{V}_{\gamma\delta}, \mathbf{V}_{\delta\delta}$ of $\mathbf{F}^{-1}$ and expand $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$: $\mathbf{a}_i' \mathbf{F}^{-1} \mathbf{a}_i = \mathbf{x}_i' \mathbf{V}_{\gamma\gamma} \mathbf{x}_i + 2\mathbf{x}_i' (\mathbf{V}_{\gamma\delta} \mathbf{w}_{j(i)}) + \mathbf{w}_{j(i)}' \mathbf{V}_{\delta\delta} \mathbf{w}_{j(i)}$. Precomputing $\mathbf{c}_j = \mathbf{V}_{\gamma\delta} \mathbf{w}_j$ ($O(Jkp)$) and $d_j = \mathbf{w}_j' \mathbf{V}_{\delta\delta} \mathbf{w}_j$ ($O(Jk^2)$) leaves per-datum cost $O(p^2 + p)$, giving the stated totals; $b'''/2b''$ equals $1/2$ for Poisson and $(1 - 2\mu_i)/2$ for the logit by direct computation. The quasi-scoring iteration and its properties are those of Kosmidis and Firth (2009); finiteness of the penalized maximizer for binomial responses is Theorem 1 of Heinze and Schemper (2002). $\square$

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